

Possible and Necessary Popularity in House Allocation PseudoCode

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1 Hypothesis

- Agents give their partial strict linear order (pairwise comparison).
- There are as many objects as agents.
- We consider only strict linear order.

2 Properties

Proposition 1 (Verify Possibly popular matching). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $\exists o \in O$ such that $o \succ_j \sigma(j)$ then return False (the matching is not popular).*

Proposition 2 (Find Possibly popular completion). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ for $o \in O$ such that $(\sigma(j), o) \notin P$ add $(\sigma(j), o)$ in $\succ_j$*

Proposition 3 (Find Possibly popular completion - Rewriting). *For a given possibly popular matching on a partial profile P , there is a complete completion that can be constructed as follows: If $\exists i, j \in N$ such that*

- $\mu(j)P_i\mu(i)$ and for $o \in O \setminus \{\mu(j)\}$ such that $(\mu(j), o) \notin P$ add $(\mu(j), o)$ in $\succ_j$
- $(\mu(i), \mu(j)) \notin P_i$, add $(\mu(i), \mu(j))$ in $\succ_i$

Proposition 4 (Find Necessarily popular matching). *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $o \in O$*

- $oP_j\sigma(j)$ or
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

Proposition 5 (Find Necessarily popular completion). *If $\exists i, j \in N$ such that $(\sigma(i), \sigma(j)), (\sigma(j), \sigma(i)) \notin P_i$ and $o \in O$ such that*

- $oP_j\sigma(j)$
- $(\sigma(j), o) \notin P_j$

Return False (it is not a necessarily popular matching).

3 Algorithms

3.1 Implementation with pairwise comparison

Algorithm 1 VerifPossiblyPopularMatching

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1: Input:  $O$  the set of  $n$  objects,  $N$  the set of  $n$  agents,  $P = (P_1, \dots, P_n)$  the partial
   profile such that  $P_i = [(o_l, o_k), \dots]$  is the list of pairwise partial preference of agent  $i$ ,
    $\mu = \{1: \sigma(1), \dots, n: \sigma(n)\}$  is the matching ( $\sigma(i)$  is the assigned object to  $i$  in the matching).
2: Output: boolean indicating either the matching  $\mu$  is possibly popular
3: for each agent  $i \in N$  do
4:   for each preference  $(o_{sup}, o_{inf}) \in P_i$  do
5:     NotFpost  $\leftarrow o_{inf}$  is  $\sigma(i)$ 
6:     NotDominatedByFpost  $\leftarrow False$ 
7:     for each preference  $(o, o') \in P_j$  do
8:       NotDominatedByFpost  $\leftarrow o_{sup}$  is  $o'$ 
9:     end for
10:    if NotFpost and NotDominatedByFpost then
11:      return False
12:    end if
13:  end for
14: end for
15: return True

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Algorithm 2 FindPossiblyPopularMatching

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1: Input:  $O$  the set of  $n$  objects,  $N$  the set of  $n$  agents,  $P = (P_1, \dots, P_n)$  the partial
   profile such that  $P_i = [(o_l, o_k), \dots]$  is the list of pairwise partial preference of agent  $i$ ,
    $\mu = \{1: \sigma(1), \dots, n: \sigma(n)\}$  is the perfect matching ( $\sigma(i)$  is the assigned object to  $i$  in the
   matching).
2: Output: A complete completion that making the matching popular.
3:  $\succ$  the complete completion  $\leftarrow$  copy of the partial completion  $P$ 
4: for each agent  $i \in N$  do
5:   for each objects  $sigma(j)$  and agents  $j$  matched  $\in \mu$  do
6:     if  $\exists(\sigma(j), \sigma(i)) \in P_i$  then
7:       for each objects  $o \in O \setminus \{\sigma(j)\}$  do
8:         if  $(\sigma(j), o) \notin P_j$  then
9:           Add  $(\sigma(j), o)$  in  $\succ_j$ 
10:        end if
11:      end for
12:    else if  $\nexists(\sigma(i), \sigma(j)) \in P_i$  then
13:      Add  $(\sigma(i), \sigma(j))$  in  $\succ_i$ 
14:    end if
15:  end for
16: end for
17: return  $\succ$ 

```

3.2 First implementation with positional ranking

Algorithm 3 VerifPossiblePopularMatching

```

1: Input:  $O$  the set of  $n$  objects,  $N$  the set of  $n$  agents,  $P = (P_1, \dots, P_n)$  the partial profile
   such that  $P_i = [?, \sigma(i), o, \dots]$  is the list of partial preference of agent  $i$  ( $?$  indicates unknown
   preference and  $\sigma(i)$  is the assigned object to  $i$  in the matching),  $\mu = \{1: \sigma(1), \dots, n: \sigma(n)\}$  is
   the matching.
2: Output: boolean indicating either the matching  $\mu$  is possibly popular,  $P$  a completion of the
   partial profile making  $\mu$  possibly popular.
3:
4: for each agent  $i \in N$  do
5:   for each preference  $p \in P_i$  do
6:     if  $p$  is unknown (equal to  $?$ ) then
7:       // Is there a completion such as  $p$  is the fpost matched to another agent in  $\mu$ ?
8:       if  $\sigma(i)$  not in  $P_i$  then
9:         Assign  $\sigma(i)$  to  $p$ 
10:      else
11:        Find a matched fpost (should not be a fpost for more than half agents) and
        assigned it to  $p$ 
12:      end if
13:    else if  $p$  is  $\sigma(i)$  then
14:      Complete the preference list  $P_i$  with the remaining objects.
15:      Break
16:    else
17:      //  $p$  is an object with a better rank than  $\sigma(i)$ 
18:      if  $p$  is a fpost matched to another agent then
19:        Continue
20:      else
21:        //  $p$  is not a matched fpost, the matching  $\mu$  cannot be popular
22:        return False,  $P$ 
23:      end if
24:    end if
25:  end for
26: end for
27: return result

```

Algorithm 4 VerifyNecessarilyPopularMatching

```
1: Input:  $O, N, P = (P_1, \dots, P_n), \mu = \{(1, \sigma(1)), \dots, (n, \sigma(n))\}$ 
2: Output: boolean indicating either the matching is necessarily popular or not.
3:  $cpt_{fpost} := 0$ 
4: for each  $i \in N$  do
5:    $Isfpost := True$ 
6:   // A matching is popular if each agent obtained either its fpost or spost.
7:   // A fpost and/or a spost can not be specified by an agent at the beginning. So we
   check if all the possible completion respect this condition.
8:   while There exist a completion of the agent's  $i$  preference list do
9:      $O_i :=$  the set of non ranked item by agent  $i$ .
10:    for Each unknown  $p \in P_i$  do
11:      Choose an  $o$  in  $O_i$  for each unknown  $p$ 
12:       $O_i \leftarrow O_i \setminus \{o\}$ 
13:    end for
14:    // At the end of this for loop, we get a completion of the agent's  $i$  preference list
15:    // Checking this completion respect the necessarily popular conditions
16:    if not  $(\sigma(i) \in \{f(i), s(i)\})$  then
17:      return False;
18:    else if  $\exists x \in P_i$  such that  $x \succ_i \sigma(i)$  and  $x$  not a fpost then
19:      return False;
20:    end if
21:    if  $\sigma(i)$  is not a fpost in this completion then
22:       $Isfpost \leftarrow False$ 
23:    end if
24:  end while
25:  if  $Isfpost$  then
26:     $cpt_{fpost} \leftarrow cpt_{fpost} + 1$ 
27:  end if
28:  // If at least half of fpost are assigned, the matching  $\mu$  is necessarily popular
29:  if  $cpt_{fpost} \geq \lceil n/2 \rceil$  then
30:    return False;
31:  end if
32: end for
33: return True;
```

4 Examples

Instance for possibly popular matching Let H an instance of the House Allocation problem such as:

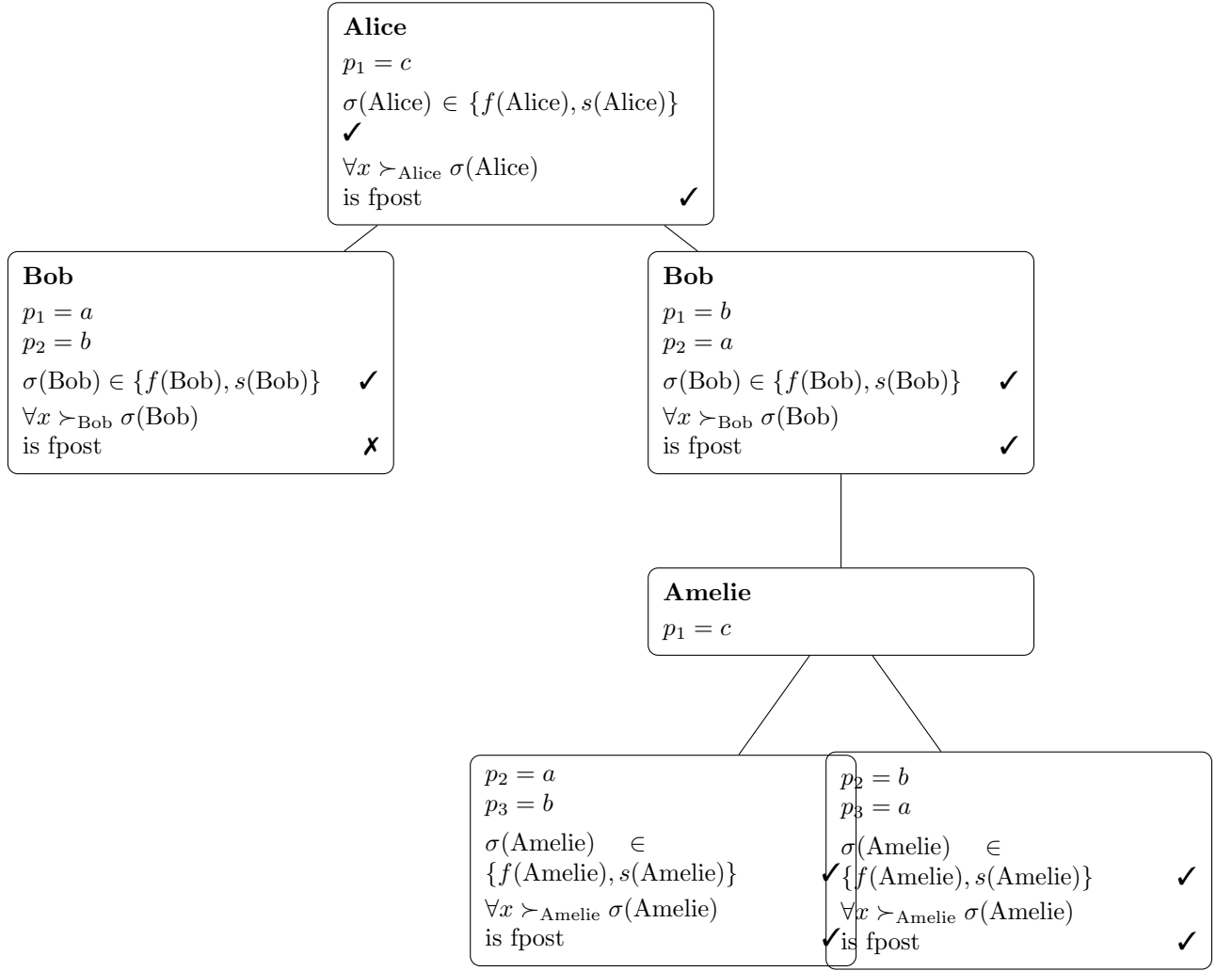
- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i .
- $\mu = \{(Alice, b), (Bob, c), (Amelie, a)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	Completions		
Alice	? a ?	$\boxed{b} \textcircled{a} c$	$\boxed{b} a c$	$c a \boxed{b}$
Bob	b ? ?	$\textcircled{b} \boxed{c} a$	$b \boxed{c} a$	$b \boxed{c} a$
Amelie	? a ?	$\textcircled{c} \boxed{a} b$	$b \boxed{a} a$	$b \boxed{a} c$
		not popular	no solution	not popular

Instance for necessarily popular matching - Non necessarily popular matching Let H an instance of the House Allocation problem such as:

- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i .
- $\mu = \{(Alice, a), (Bob, b), (Amelie, c)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	Completions		
Alice	? a b	$\boxed{b} \textcircled{a} c$	$\boxed{b} a c$	$c a \boxed{b}$
Bob	? ? a	$\textcircled{b} \boxed{c} a$	$b \boxed{c} a$	$b \boxed{c} a$
Amelie	c ? ?	$\textcircled{c} \boxed{a} b$	$b \boxed{a} a$	$b \boxed{a} c$
		not popular	no solution	not popular



5 Proof

5.1 Proof Verif Possible Popular Matching

Hypotheses

- N is a set of n agents and O a set of n objects.
- We assume a **perfect matching**: $|N| = |O|$ and all agents/objects are matched.
- Preferences are given as **partial orders** (pairwise comparisons).
- We consider **completions of the partial preference profile**.
- μ is a matching.

Definitions

- A **completion** is a strict complete order extending the partial preferences.
- A matching μ is **popular** if for every matching μ' :

$$|\{i \in N : \mu(i) \succ_i \mu'(i)\}| \geq |\{i \in N : \mu'(i) \succ_i \mu(i)\}|.$$

- μ is **possibly popular** if it is popular under at least one completion of the partial preference profile.

Proof of $\Rightarrow$ Suppose that there exist $i, j \in N$ such that

$$\sigma(j) P_i \sigma(i) \quad \text{and} \quad \exists o \in O \text{ such that } o \succ_j \sigma(j).$$

We prove that μ is not possibly popular.

Let $\succ$ be any completion of the partial preference profile P . We show that μ is not popular under $\succ$ by constructing a matching μ' that defeats μ .

Construction of μ'

- If o is unmatched in μ , define:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o.$$

- Otherwise, there exists $k \in N$ such that $o = \sigma(k)$. Define a local exchange:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o, \quad \mu'(k) = \sigma(i),$$

and for all $\ell \notin \{i, j, k\}$:

$$\mu'(\ell) = \mu(\ell).$$

Validity

The matching μ' is valid since it corresponds to a permutation of objects among at most three agents and leaves all others unchanged.

Comparison of μ and μ'

- Agent i prefers μ' :

$$\sigma(j) \succ_i \sigma(i).$$

- Agent j prefers μ' :

$$o \succ_j \sigma(j).$$

- At most one agent (namely k if it exists) may prefer μ .
- All other agents are indifferent.

Vote count

$$|\{i \in N : \mu'(i) \succ_i \mu(i)\}| \geq 2 \quad \text{and} \quad |\{i \in N : \mu(i) \succ_i \mu'(i)\}| \leq 1.$$

Thus, μ' is more popular than μ , contradicting the popularity of μ .

Since this holds for any completion, μ is not possibly popular. □

Proof of $\Leftarrow$ Suppose that μ is not possibly popular. Then for every completion of the partial preference profile, μ is not popular.

We use the following characterization from Abraham et al.:

Lemma (Abraham et al.) A matching μ is popular if and only if:

- every agent is matched to either its *fpost* or its *spost*,
- and every *fpost* is matched.

By contraposition of the lemma (Abraham et al), since μ is not popular for any completion, for every completion there exists at least one agent i such that:

$$\mu(i) \notin \{fpost(i), spost(i)\}.$$

In particular, this implies:

$$fpost(i) \succ_i \mu(i).$$

Let $j \in N$ be such that:

$$\mu(j) = fpost(i).$$

Thus:

$$\sigma(j) \succ_i \sigma(i).$$

Let $k \in N$ be such that:

$$\mu(k) = spost(i).$$

We distinguish the following cases:

1. If $\mu(j) \notin \{fpost(j), spost(j)\}$, then by definition of *fpost*(j) there exists an object $o \in O$ such that:

$$o \succ_j \mu(j),$$

and the property holds.

2. If $\mu(j) = spost(j)$, then by definition of *fpost*(j) we have:

$$fpost(j) \succ_j \mu(j),$$

so there exists $o = fpost(j)$ such that:

$$o \succ_j \sigma(j).$$

3. If $\sigma(j) = fpost(j)$ and $\sigma(k) = fpost(k)$, then both agents receive their *fpost*.

Since μ is assumed not popular, this contradicts the characterization of Abraham et al., which requires that every agent is matched to either its *fpost* or *spost*, and that every *fpost* is matched. Hence, this situation cannot occur under our assumption.

In all cases, we obtain:

$$\exists o \in O \text{ such that } o \succ_j \sigma(j).$$

Therefore:

$$\exists i, j \in N \text{ such that } \sigma(j) \succ_i \sigma(i) \quad \text{and} \quad \exists o \in O \text{ such that } o \succ_j \sigma(j).$$

This proves the desired property. □

5.2 Proof Find Possibly Popular Completion

Hypotheses

- N is a set of n agents and O a set of n objects.
- We assume a **perfect matching**: $|N| = |O|$ and all agents/objects are matched.
- Preferences are given as **partial orders** (pairwise comparisons).
- We consider **completions of the partial preference profile** $P \succ$.
- μ is a possibly popular matching.

Definitions

- $\forall i \in N, \forall o \in O \setminus \{\sigma(i)\}, \exists(\sigma(i), o) \text{ or } (o, \sigma(i)) \text{ in } \succ_i$.
- **From Handbook of Computational Social Choice (p26)**: $a \succsim b$ or $b \succsim a$ for all $a \neq b \in O$ and any voter $i \in N$
- **Szpilrajn's Theorem**: Any partial order extends to a total order.

Proof From the definition of a possibly popular matching μ , we know that there exists a completion $\succ$ of the partial preference profile P such that μ is popular under $\succ$. Under this completion and from the Abraham et al. characterization of popular matchings, we have that every agent is matched to either its *fpost* or its *spost*, and every *fpost* is matched. Hence:

1. Because there are as many agents as houses, for all agent $i \in N$, $\mu(i)$ is a *fpost* of any agent j such that $fpost(j) \succ_i o$ for all houses $o \in O \setminus \{fpost(j)\}$.
2. For all $\mu(i)$, if there does not exist an agent j such that $\mu(i) = fpost(j)$, then μ is not popular under $\succ$, which is a contradiction of the second condition of the characterization of Abraham et al.
3. For all $\mu(i)$, if $\mu(i) \succeq_i spost(i)$ then the first condition of the characterization of Abraham et al. is verified. In the other case, when $spost(i) \succ_i \mu(i)$, the first condition is not verified and μ is not popular under $\succ$.

Reasoning with these three points, we are looking to reconstruct a completion $\succ$ from the partial preference profile P .

Let's prove that the construction is well-defined by showing that:

1. The obtained completion is composed of strict linear orders over the set of all houses and it is irreflexive (no ties), asymmetric and transitive.
2. The obtained completion is a total linear order. Each agent i has a complete preference list over all houses O , i.e. for all agent $i \in N$ there exists a relation $\succ_i$ such that for all $o, o' \in O$, either $o \succ_i o'$ or $o' \succ_i o$.

3. The obtained order is transitive. For all agent $i \in N$, for all $a, b, c \in O$, if $a \succ_i b$ and $b \succ_i c$ then $a \succ_i c$.
4. The completion should be consistent with the partial preference profile P .
5. The completion does not have any cycle. For all agent $i \in N$, for all $a, b, c \in O$, if $a \succ_i b$ and $b \succ_i c$ then not $c \succ_i a$.

1. Strict linear order

We work with *strict* linear orders, consistently with the partial preference profile P .

Irreflexivity means that no house is in relation with itself, i.e. for all $i \in N$ and for all $o \in O$, $(o, o) \notin succ_i$. For all $i \in N$ and for all $o, o' \in O$ such that $(o', o) \notin P_i$: if $o = o'$, then $(o, o') \notin \succ_i$ because our construction rules do not add any relation between o and itself. Otherwise the irreflexivity is verified because the relation is not between o and itself.

Asymmetry does not allow a house o preferred to another o' to be also less preferred than o' , i.e. for all $i \in N$ and for all $o, o' \in O$ ($o \neq o'$) such that $(o, o') \in \succ_i$ then $(o', o) \notin \succ_i$. if $o = \sigma(i)$ and $o' = \sigma(j)$ and $(\sigma(i), \sigma(j)) \notin P_i$, then $(o, o') \in \succ_i$. By contradiction, let's suppose that $(\sigma(j), \sigma(i)) \in \succ_i$.

=> **Need to rewrite the proposition!**

Sources:

- <https://fr.wikipedia.org/wiki/Th>